# READING NOTES

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## Contents

## 1. Introduction

The motivation behind this running record of reading notes on mathematics is to gather the ongoing observations, ideas and questions which arise during my study. The entries are primarily for my own understanding, but I hope they may also be of interest to others who enjoy exploring mathematics. As my understanding evolves, some notes may be revised in the future.

## 2. Notes

**13th Aug, 2026.** In algebaric geometry, given an exact sequence of presheaves

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \cdots \longrightarrow \mathcal{F}_n \longrightarrow,$$

we show that the following sequence

$$0 \longrightarrow \mathcal{F}_1 U \longrightarrow \mathcal{F}_2 U \longrightarrow \cdots \longrightarrow \mathcal{F}_n U \longrightarrow,$$

is also exact. Well, this is an iff result and given the definition of a presheaf, which is defined locally, the first exact sequence gives us the second one. We work the other way around by, again, using the definition of presheaves.

**12th Aug, 2026.** Mathematical ventures on this day included some more reading into scheme-theoretic algebraic notions from Eisenbud, along with some revisitation of linear algebra, particularly diagonal matrices and diagonalizability, and also a result in metric spaces.

A diagonal matrix is a matrix whose entries away from the main diagonal are all zero, i.e.

$$D = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}.$$

It is interesting that a matrix $A$ is diagonalizable precisely when there is a basis consisting of eigenvectors of $A$, equivalently when

$$P^{-1}AP = D$$

for some invertible matrix $P$ and diagonal matrix $D$.

The columns of $P$ are simply the linearly independent eigenvectors of $A$. If

$$Av_i = \lambda_i v_i, \qquad i = 1, \ldots, n,$$

then we take

$$P = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}, \qquad D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

Since the eigenvectors are linearly independent, $P$ is invertible, and from

$$AP = PD$$

we obtain

$$P^{-1}AP = D.$$

So, essentially, determining $P$ amounts to finding a basis of eigenvectors of $A$ and placing these eigenvectors as the columns of $P$.

In metric spaces, every finite metric space with $n$ points can be isometrically embedded into $\mathbb{R}^{n-1}$ with the usual Euclidean metric. It is quite interesting to see an abstract finite metric space being realised concretely as a configuration of points in Euclidean space.

Alongside this, $\operatorname{Spec}(R)$ both as the set of prime ideals of $R$ and through the geometric structure given by the Zariski topology is quite interesting.

We can think of

$$\operatorname{Spec}(R) = \{\mathfrak{p} \subseteq R : \mathfrak{p} \text{ is prime}\}$$

as a collection of prime ideals, and also as a geometric object by putting the Zariski topology on it. For an ideal $I \subseteq R$, we consider

$$V(I) = \{\mathfrak{p} \in \operatorname{Spec}(R) : I \subseteq \mathfrak{p}\},$$

and these sets form the closed sets of the Zariski topology. There is also a structure sheaf $\mathcal{O}_{\operatorname{Spec}(R)}$, and together with the underlying topological space, this gives us the affine scheme associated to $R$.

**10th Aug, 2026.** Mathematics on this day included some interesting results about groups acting on metric spaces. One such result was that if a group $G$ acts freely by isometries on $\mathbb{R}^n$, equipped with the Euclidean metric, then $G$ must be torsion-free. This is quite interesting since the geometry of the space places an algebraic restriction on the group itself.

In particular, the idea of algebraic sets as zero sets of ideals and the way varieties arise from imposing irreducibility on these sets is interesting.

**09th Aug, 2026.** Mathematical ventures on this day included some revisits to linear algebra and analysis, for the most part. The theory of matrices is very rich and gives us some very strong results. For instance, the eigenvalues of a matrix whose entries are non-negative and whose row sums are all equal to 1 must all have absolute value at most 1. That is, if $A = a_{ij}$ satisfies

$$a_{ij} \geq 0, \qquad \sum_j a_{ij} = 1,$$

then every eigenvalue $\lambda$ of $A$ satisfies

$$|\lambda| \leq 1.$$

In analysis, the Bolzano-Weierstrass theorem tells us that every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence, extending that the *Arzelà-Ascoli* theorem tells us that under suitable conditions, a sequence of functions has a *uniformly* convergent subsequence. In particular, *a uniformly bounded and equicontinuous family of continuous functions on a compact space is relatively compact in the uniform topology.*

Also the idea that a finite metric space can be isometrically embedded into some Euclidean space $\mathbb{R}^n$ is particularly interesting. Thus, given a finite metric space $X$, there exists some $n$ and a distance-preserving map

$$f : X \longrightarrow \mathbb{R}^n.$$

**08th Aug, 2026.** Mathematics on this day included some readings on the realisation of groups as metric spaces. We start by taking a group $G$ and its generating set $S$. Now, for any two elements $g_1, g_2 \in G$ we define the *distance* between them to be the word length $g_1^{-1}g_2$ with respect to $S$. This induces a metric on $G$, refered to as the *word metric.* Well, to be precise, the word metric is defined as

$$d_S(g_1, g_2) = |g_1^{-1}g_2|_S,$$

where $|g_1^{-1}g_2|_S$ denotes the minimum number of generators from $S \cup S^{-1}$ needed to express $g_1^{-1}g_2$. Moreover, when we think of groups in terms of their Cayley graaphs, this word metric is precisely the *path metric* of the graph.

Another interesting thing about viewing groups as metric spaces is the analogue of isomorphisms which we obtain in the latter, called an *isometry.* An isometry is simply a bijective function, say $f$ defined as:

$$f : (X, d_X) \to (Y, d_Y),$$

such that

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2)$$

for all pairs $x_1, x_2 \in X$.

Another interesting notion is that of the group of symmetries of a metric space $X$, denoted as $\mathrm{Isom}(X)$. And a group action of a group $G$ on a metric space $X$ would then be the following group homomorphism

$$G \to \mathrm{Isom}(X).$$

So essentially, for each element $g \in G$, the map gives us an isometry $X \to X$.

**07th Aug, 2026.** Some mathematics discussions[1] on this day included an alternate proof of the *Bolzano–Weierstrass theorem*. The idea was to start with a bounded sequence $(a_n) \subseteq [a, b]$, divide the interval into two parts, and keep the part which contains infinitely many terms of the sequence. One can keep doing this, obtaining nested intervals

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$$

whose lengths tend to zero. One can then choose a subsequence by taking a term from each of these intervals, and this subsequence will converge to the unique point determined by the nested intervals. I particularly liked the rather physical analogy used to explain this.

Also, the *Archimedean property* can be realised through the Tortoise and Hare story, and the *Monotone Convergence Theorem* can be visualised by repeatedly partitioning a bounded interval and narrowing down towards a certain point.

In another news, an algebraic set can be defined using an ideal rather than a collection of polynomials. Given an ideal $\mathfrak{I} \subseteq k[x_1, \ldots, x_n]$, we can consider

$$V(\mathfrak{I}) = \{a \in \mathbb{A}_k^n : f(a) = 0 \text{ for every } f \in \mathfrak{I}\}$$

.Moreover, the radical of an ideal has some interesting properties that one can play around with in commutative algebra. In particular,

$$V(\mathfrak{I}) = V(\sqrt{\mathfrak{I}}),$$

which is quite interesting since the algebraic set cannot distinguish between an ideal and its radical.

Modules, on the other hand, generally have two ways of being viewed. While I prefer the ring-action approach,

$$R \times M \longrightarrow M,$$

I hope to learn to appreciate the linear algebra approach as well, where modules over rings are considered analogous to vector spaces over a field.

**05th Aug, 2026.** One of the more surprising ideas I encountered today was that the common notion of the zero set of a collection of polynomials is elevated to a central geometric object in algebraic geometry. Such a set is called an *algebraic set*, while a *variety* is, roughly speaking, an irreducible algebraic set (depending on the convention being used).

I also found it interesting that these objects are studied in affine space ($\mathbb{A}_k^n$). Although affine space has points with coordinates such as $(0, 0)$ or, more generally, $(0, \ldots, 0)$, it is

[1] Please refer to this webpage and the notes provided there for these ideas.

not regarded as having a distinguished origin in the geometric sense. The point $(0, \dots, 0)$ is then simply another point of the space, and whether it belongs to the zero set of a polynomial depends entirely on the polynomial itself. For instance, if the constant term is nonzero, then $(0, \dots, 0)$ need not be a zero of that polynomial.

Another pleasing observation was that the Cayley graph of the free group $F_2$, with respect to its standard free generators, is the infinite 4-regular tree $T_4$. Starting from any vertex, the tree branches outward in a remarkably symmetric fashion, and after the initial four edges, there are $4 \cdot 3^{n-1}$ new vertices at distance $n$ from the starting point.

Moreover, algebraic sets are interesting objects. Although they can be defined over any field, many of their fundamental properties are studied over algebraically closed fields. There is also the *Zariski topology*, which arises naturally on affine spaces and whose closed sets are precisely the algebraic sets, at least for $n = 1$ and $n = 2$.

**04th Aug, 2026.** Well, the Cayley graph of a free group with respect to a free basis is an infinite regular tree. This graph has some interesting valencies at each point and it is unlike the finite groups' graphs which contain cycles. In other news, $SL_2(\mathbb{Z})$ has some interesting elements which are also its generators. It would be interesting to explore the geometry of its Cayley graph.

**03rd Aug, 2026.** Today I learned that automorphisms of a group induce automorphisms of its Cayley graph (provided the generating set is preserved). This is interesting because every automorphism of a group is a permutation of its underlying set, so there is a natural embedding $\mathrm{Aut}(G) \hookrightarrow \mathrm{Sym}(G)$. Thus, the same automorphism acts both as a permutation of the group and as a symmetry of its Cayley graph.